(Q1)

$P(E = High)$

Setting up the sum.

→ Start from factorization and sum out every variable except E:

$P(E=H) = \Sigma_{\{W,D,O,A,L\}} P(W)\ P(D)\ P(O|D)\ P(W|O)\ P(A|W,O)\ P(L|O)\ P(E=H|A,L)$

Put each sum as far inward as it will go:

$P(E=H) = \Sigma_{\{A,L\}} P(E=H|A,L) \cdot P(A,L)$

$P(A,L) = \Sigma_O P(O) \cdot P(A|O) \cdot P(L|O)$

(A & L are independent given O)

where, $P(O) = \Sigma_D P(D)\ P(O|D)$

$P(A|O) = \Sigma_W P(W)\ P(A|W,O)$

STEP 1: ~~$P(A=ON|O)$~~ $P(O)$

$P(ON|High) = P(Wd) \cdot P(High|Wd) \cdot P(We) \cdot P(High|We)$

$= 0.7(0.45) + 0.3(0.8)$

$= 0.555$

$P(O = LOW) = 1 - 0.55 = 0.45$

STEP 2: $P(A=ON|O)$

$P(ON|High) = 0.5(0.9) + (0.3)(0.4) + (0.2)(0.1)$

$= 0.59$

$P(OFF|High) = 1 - 0.59 = 0.41$

$P(ON|Low) = 0.5(0.6) + 0.3(0.15) + 0.2(0.05)$

$= 0.35$

$P(OFF|Low) = 1 - 0.35 = 0.65$

STEP 3: $P(L|O)$

$P(Heavy|High) = 0.75$

$P(Heavy|Low) = 0.35$

$P(Light|High) = 0.25$

$P(Light|Low) = 0.65$

STEP 4: $P(A,L) = \Sigma_O P(O) P(A|O) P(L|O)$

$P(On, Heavy) = 0.55(0.59)(0.75) + 0.44(0.35)(0.35)$

$= 0.30$

$P(On, Light) = 0.55(0.59)(0.25) + 0.44(0.35)(0.65)$

$= 0.18$

$P(Off, Heavy) = 0.55(0.41)(0.75) + (0.44)(0.64)(0.35)$

$= 0.27$

$P(Off, Light) = 0.55(0.41)(0.25) + 0.44(0.64)(0.65)$

$= 0.24$

CHECK: $0.3 + 0.18 + 0.27 + 0.24 \simeq 1$

STEP 5:

$P(E=High) = \sum_{[A,L]} P(E=H \mid A, L) \cdot P(A, L)$

A	L	$P(E=H \mid A, L)$	$P(A, L)$	Product
ON	Heavy	0.92	0.3	0.28
ON	Light	0.68	0.18	0.12
Off	Heavy	0.55	0.27	0.15
Off	Light	0.12	0.24	0.03
			SUM	0.57

$P(E=High) \approx 0.58$

This matches the code output.